

ADVANCED PERSISTENT THREATS

WHAT, WHO, HOW, WHEN, WHERE, AND WHY?

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WHAT IS AN APT?

Advanced Persistent Threats (APTs) are malicious cyber operations that, as their name suggests, possess three defining characteristics:

- They represent a **threat** because they pursue highly specific objectives, typically involving the theft of classified or critical information, financial gain, or large-scale sabotage.
- They are **persistent** because the malware is specifically designed to remain hidden within the target system. It avoids raising suspicion during installation, conceals its communications with external servers through strong encryption, and transmits small amounts of data at regular intervals. This enables it to remain inside the victim's network for as long as possible. Once its objectives have been achieved, it will often attempt to uninstall itself, removing logs and other traces that could reveal its origin, infection method, activities, or any other identifying information.
- They are **advanced** because they employ sophisticated techniques such as encrypted communication channels, zero-day vulnerabilities, and advanced evasion and propagation capabilities to bypass security defenses and compromise additional systems.

The acronym **APT** is also commonly used not only to describe the malicious software itself, but also the type of attack and the threat actor or group responsible for developing and operating it. We will discuss these groups later in this article.

WHO IS BEHIND AN ADVANCED PERSISTENT THREAT?

Unfortunately, this is one of the most difficult questions to answer. It is virtually impossible to determine with absolute certainty who is responsible for an APT campaign. Analysts can only examine the available evidence and infer who may have developed or deployed the malware. For this reason, cybersecurity professionals speak of **attribution** rather than definitive identification.

Malware analysts working for organizations such as CERTs (Computer Emergency Response Teams) or antivirus companies usually assign names both to the malware and to the threat group behind it. These names may be distinctive—such as **Turla** or

Carbanak—or simply follow a numerical convention such as **APTXX**, where *XX* represents a number.

These groups are known to possess significant human, financial, and technical resources. Without such capabilities, they would not be able to develop and operate malware of this sophistication. However, the source of their funding generally remains unknown because no organization publicly claims responsibility for their activities.

Typically, analysts attribute a group's origin based on evidence extracted from the malware itself, such as the language used in source code comments, compilation artifacts, or operational patterns previously observed in other attributed APT campaigns.

Does this mean we can confidently conclude that a government, corporation, or organization with sufficient financial, human, and technological resources is behind a specific APT group?

Absolutely not.

Although such organizations certainly possess the necessary capabilities, no conclusive evidence usually exists to prove or disprove their involvement.

HOW DOES AN APT OPERATE?

APT groups generally rely on **zero-day attacks**. These attacks exploit software vulnerabilities for which no security patch yet exists, either because the vulnerability is still unknown or because a fix has not yet been developed.

Another common technique is the **watering hole attack**. In this type of attack, threat actors compromise legitimate third-party websites that are frequently visited by employees of the intended target organization, such as social media platforms or industry-specific websites. The primary targets are the organization's users, who unknowingly infect their own systems simply by visiting those compromised websites.

WHEN DOES AN APT ATTACK?

According to several analyses, some malware families appear to begin and end their activities during specific time windows. This behavior may provide investigators with additional clues regarding their possible geographic origin.

Nevertheless, anyone can become infected at any time and from virtually anywhere due to the nature of these threats. Operating schedules alone are never sufficient to determine attribution.

WHERE CAN AN APT BE FOUND?

Unfortunately, there is no specific category of system that can be considered immune to APT infections.

Examples have been identified in:

- Internet of Things (IoT) devices
- Routers and networking equipment
- Nuclear power plant control systems
- Personal computers
- Smartphones

Any individual, company, or government organization can become the target of an APT group if sufficient strategic value exists. However, the most common targets are organizations with substantial financial resources or infrastructures considered essential to national security, including:

- Nuclear power plants
- Water treatment facilities
- Electric power infrastructure
- Critical industrial systems
- Government institutions

WHY ARE APTS CREATED?

The most common motivations behind the development and deployment of Advanced Persistent Threats include:

- Conducting **false-flag operations** in order to discredit rival governments or organizations.
- Sabotaging critical infrastructure to create political or economic instability.
- Stealing information for future intelligence or military operations.
- Obtaining financial gain.
- Achieving political, strategic, or military advantages.

In short, any objective capable of providing the threat actor—or the organization allegedly supporting it—with an economic, political, or strategic benefit may justify the deployment of an APT campaign.

Those interested in exploring specific APT groups or campaigns in greater depth can consult the information published by **ThaiCERT**. Their database highlights the predominance of groups attributed to **Russia** and **China**, followed by actors attributed to **Iran**, **India**, **North Korea**, **South Korea**, and **Turkey**. The United States also has numerous attributed groups, while Europe has comparatively fewer, with cases linked to

Ukraine, Romania, and France. In Latin America, **Brazil** and **Mexico** appear among the attributed countries.

At the time this article was written, no APT group had been publicly attributed to Spain.

I hope this article serves as a useful introduction to the world of cyber espionage and Advanced Persistent Threats.

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